

Long division



1 Simplify the following:

a $\frac{z + 10}{10}$

b $\frac{9x^3 + 5x^5}{x}$

c $\frac{18x^3 - 12x^2 + 24}{3}$

d $\frac{48x^4 + 12x^3 - 30x^2 - 42x}{6x}$

2 Complete the following statement:

$$\frac{11x^7 - 11x^5}{2x^1} = 3x^4 - 4x^2$$

3 What polynomial, when divided by $2x^2$, produces $8x^6 - 5x^4 + 6x^2$ as a quotient?

4 Consider the division $\frac{x^3 + 1}{x + 1}$.

What is the best way to write the polynomial $x^3 + 1$ in order to keep all of the like terms aligned during the long division process?

5 Divide $x^2 - 6x + 1$ by $x + 3$ using long division.

a State the remainder of this division.

b State the quotient of this division.

6 Perform the following using long division. State the quotient and remainder.

a $(x^2 - 8x + 10) \div (x + 4)$

b $(x^2 - 5x + 11) \div (x + 5)$

c $(x^2 + 3x + 2) \div (x - 3)$

d $(2x^3 + 7x^2 + 10x + 8) \div (x + 2)$

e $(3x^3 - 15x^2 + 2x - 10) \div (x - 5)$

f $(3x^3 + 2x^2 + 9x + 6) \div (x^2 + 3)$

g $(x^4 - 81) \div (x + 3)$

h $(4x^3 - 41x + 15) \div (x - 3)$

Synthetic division

7 Consider the division $(2x^2 + 3x - 4) \div (x + 4)$.

Is it possible to use synthetic division to perform this division?

8 Write each division in terms of the quotient and remainder by using synthetic division:



a
$$\frac{x^3 + 2x^2 - 29x - 30}{x + 6}$$

b
$$\frac{x^3 + x^2 - 12}{x - 2}$$

c
$$\frac{x^3 - 23x - 10}{x - 5}$$

d
$$\frac{3x^3 - 19x^2 + 25x - 22}{x - 5}$$

e
$$\frac{-3x^3 + 52x + 17}{x + 4}$$

f
$$\frac{-3x^3 - 12x^2 + 28}{x + 3}$$

g
$$\frac{x^4 - 2x^3 - 11x^2 - 18x - 9}{x - 5}$$

h
$$\frac{x^4 - 7x^3 + 7x^2 + 15x}{x - 3}$$

i
$$\frac{2x^4 - 3x^3 - 11x^2 + 6x}{x - 3}$$

j
$$\frac{x^4 + 9x^3 + 17x^2 - 13x + 10}{x + 5}$$

k
$$\frac{x^5 + 32}{x + 2}$$

l
$$\frac{x^7 - 128}{x - 2}$$

9 Consider $x^3 + 6x^2 + 12x + 16$.

- a Use synthetic division to divide $x^3 + 6x^2 + 12x + 16$ by $x + 4$.
- b State the polynomial $x^3 + 6x^2 + 12x + 16$ in terms of the factor $x + 4$ and the quotient.

10 Consider the statement $\frac{x^3 - x^2 - 10x + 6}{x + 2} = ax^2 + bx + c + \frac{d}{x + 2}$.

- a Use synthetic division to rewrite the division in terms of the quotient and remainder.
- b Find $a + b + c + d$.

11 For each of the following polynomial functions:

- i What does $P(5)$ represent?
- ii Use synthetic division to divide $P(x)$ by $x - 5$.
- iii Now, determine $P(5)$.

a $P(x) = x^2 - x - 20$

b $P(x) = 3x^3 - 18x^2 + 19x - 16$

12 Consider $P(x) = x^3 - x + 2$.

- a What does $P(0.5)$ represent?
- b Use synthetic division to divide $P(x)$ by $x - 0.5$.
- c Now, find $P(0.5)$.

13 Consider $P(x) = x^4 - 6x^2 + 11$.



- a What does $P(\sqrt{2})$ represent?
- b Use synthetic division to divide $P(x)$ by $x - \sqrt{2}$.
- c Now, find $P(\sqrt{2})$.

14 Consider $P(x) = x^2 + 2x - 8$.

- a Use synthetic division to divide $P(x)$ by $x + 4$.
- b Now, find $P(-4)$.

15 Consider $P(x) = x^2 - 9x + 21$.

- a Use synthetic division to divide $P(x)$ by $x - 5$.
- b Now, find $P(5)$.

16 Consider $P(x) = x^5 - 1$.

- a Use synthetic division to divide $P(x)$ by $x - 1$.
- b Now, find $P(1)$.

17 Consider $P(x) = x^3 + 4x + 2$. Use synthetic division to find $P(\sqrt[3]{3})$.

18 Consider $P(x) = 3x^3 - 10x^2 - 12x + 16$.

- a Use synthetic division to divide $P(x)$ by $x - 4$.
- b Determine whether 4 is a zero of $P(x)$.
- c Use synthetic division to divide $P(x)$ by $x - 8$.
- d Determine whether 8 is a zero of $P(x)$.

19 Consider $P(x) = x^4 - 9x^3 + 22x^2 - 13x + 15$.

- a Use synthetic division to divide $P(x)$ by $x - 5$.
- b Determine whether 5 is a zero of $P(x)$.
- c Use synthetic division to divide $P(x)$ by $x + 3$.
- d Determine whether -3 is a zero of $P(x)$.

20 Consider $P(x) = x^2 - 7x + 12$.



- a** Use synthetic division to divide $P(x)$ by $x - 3$.
- b** Determine whether 3 is a zero of $P(x)$.

21 Consider $P(x) = x^2 + 3x - 11$.

- a** Use synthetic division to divide $P(x)$ by $x + 5$.
- b** Determine whether -5 is a zero of $P(x)$.

22 Consider $P(x) = 3x^3 + 4x^2 - 12x + 10$.

- a** Use synthetic division to divide $P(x)$ by $x + 3$.
- b** Determine whether -3 is a zero of $P(x)$.

23 Consider $P(x) = 3x^3 - 3x^2 - 6x$.

- a** Use synthetic division to divide $P(x)$ by $x - 2$.
- b** Determine whether 2 is a zero of $P(x)$.

24 Consider $P(x) = 2x^3 - x^2 - x - 3$.

- a** Use synthetic division to divide $P(x)$ by $x - 1.5$.
- b** Determine whether 1.5 is a zero of $P(x)$.

25 Consider $P(x) = 8x^3 + 2x^2 + 3x - 2$.

- a** Use synthetic division to divide $P(x)$ by $x - 0.25$.
- b** Determine whether 0.25 is a zero of $P(x)$.

26 Consider $P(x) = 3x^6 - x^4 + 2x^2 - 24$.

- a** Use synthetic division to divide $P(x)$ by $x - \sqrt{2}$.
- b** Determine whether $\sqrt{2}$ is a zero of $P(x)$.

27 Consider $2x^3 - x^2 - 9x - 4 = 0$.



- a Write down all the possible rational solutions.
- b Determine whether $x = \frac{1}{2}$ is a solution of the equation.
- c Determine whether $x = -\frac{1}{2}$ is a solution of the equation.
- d Rewrite $2x^3 - x^2 - 9x - 4$ in the form $(Ax + B)(Cx^2 + Dx + E)$.
- e Now, find all the solutions of $2x^3 - x^2 - 9x - 4 = 0$.

28 Consider $f(x) = x^3 - 6x^2 + 11x - 6$.

- a Write down all the possible rational zeros.
- b Determine whether $x = -1$ is a zero of $f(x)$.
- c Determine whether $x = 1$ is a zero of $f(x)$.
- d Factor $x^3 - 6x^2 + 11x - 6$ completely.
- e Now, find all the zeros of the polynomial $f(x)$.

29 Consider $P(x) = 6x^3 - 11x^2 + 6x - 1$ and the table of values for $P(x)$ on the integers in the interval $0 \leq x \leq 5$ as shown:

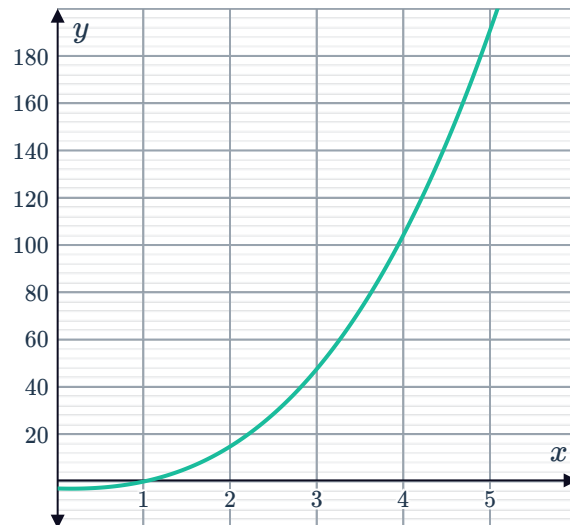
x	0	1	2	3	4	5
$P(x)$	-1	0	15	80	231	504

- a Use the table to find a solution to the equation $6x^3 - 11x^2 + 6x - 1 = 0$.
- b Use synthetic division to divide $P(x)$ by $x - 1$.
- c Determine whether 1 is a zero of $P(x)$.
- d Now, find all the solutions of the equation $6x^3 - 11x^2 + 6x - 1 = 0$.

30 Consider $P(x) = x^3 + 3x^2 - x - 3$ and the graph of $y = P(x)$ shown:

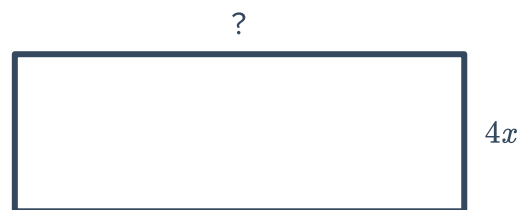


- a Use the graph to find a solution to the equation $x^3 + 3x^2 - x - 3 = 0$.
- b Use synthetic division to divide $P(x)$ by $x - 1$.
- c Determine whether 1 is a root of $P(x)$.
- d Factor $x^3 + 3x^2 - x - 3$ completely.
- e Now, find all the solutions of the equation $x^3 + 3x^2 - x - 3 = 0$.

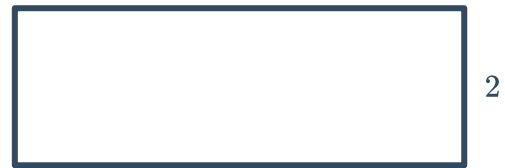


Applications

- 31 The rectangle has an area of $(0.2x^3 + 0.6x^2 + 0.45x + 0.1)$ units² and width of $(x + 0.5)$ units.
 - a Find a rational expression for the length of the base.
 - b Simplify the expression for the length of the rectangle.
- 32 The area of a parallelogram is given by $(x^4 - 7x^2 + 4x - 6)$ in², and its base measures $(x + 3)$ in in length. Find the perpendicular height of the parallelogram.
- 33 The rectangle shown has an area of $(4x^4 - 8x)$ units². Find a polynomial expression for its length.



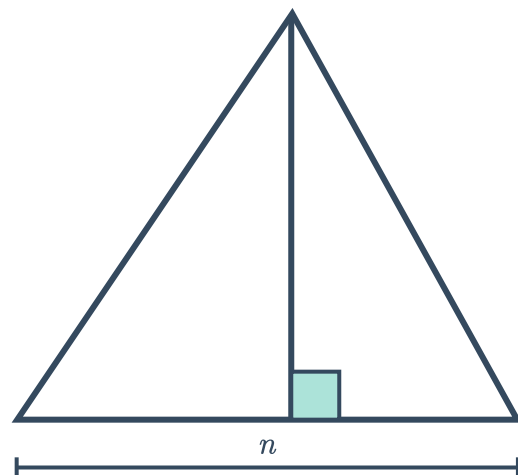
- 34** The rectangle shown has an area of $(4x^4 - 8x)$ units².
Find a polynomial expression for its length.



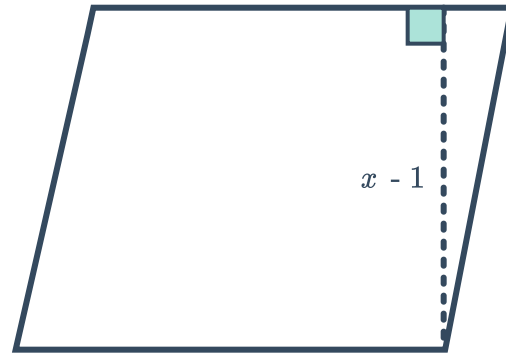
- 35** The rectangle shown has an area of $(5x^3 + 7x^2 - 18x - 8)$ units².
Find a polynomial expression for its length.



- 36** The triangle shown has an area of $(13n^3 + 11n^2 + 29n)$ units².
Find a polynomial expression for the perpendicular height.



- 37** This parallelogram has an area of $(2x^3 + 4x^2 - 5x - 1)$ units².
Find a polynomial expression for the length of its base.



- 38** If the distance traveled is $(2x^3 - 2x^2 + 5x + 9)$ mi and the speed is $(x + 1)$ mi/h , find the time traveled in hours.
- 39** It costs $(4x^5 + 5x^4 + 2x^3 + 10x^2 - 5x + 14)$ dollars to replace the lawn in the backyard. If the new lawn costs $(x + 2)$ dollars per square meter, find the area of the lawn.
- 40** The amount of dust, in milligrams, collected on a table after x days without dusting is modeled by the function $f(x) = 9x^3 - 30x^2 - 34x + 60$. After how many days without dusting will the table have collected 20 mg of dust?